

PROMPT ENGINEERING

1. Summarising Notes Prompt

Summarize the following notes in simple language. Highlight the main points and create bullet points for quick revision.

AI Response

Artificial Intelligence (AI) is the simulation of human intelligence by machines. It enables computers to learn, reason, and solve problems. AI is widely used in healthcare, education, transportation, and business. Machine Learning is a branch of AI that learns from data, while Generative AI creates new content such as text and images.

Key Points:

AI mimics human intelligence.

Machine Learning learns from data.

Generative AI creates content.

AI is used in many industries.

Why This Prompt Worked

This prompt clearly states the task and requests a simple summary with bullet points. The instructions help the AI generate an organized and easy-to-understand response.

2. Generating a Study Plan

Prompt

Create a 7-day study plan for a first-year engineering student preparing for a Python programming examination. Include daily topics and revision time.

AI Response

Day 1: Introduction to Python and Variables

Day 2: Data Types and Operators

Day 3: Conditional Statements (if, else, elif)

Day 4: Loops and Functions

Day 5: Lists, Tuples, Dictionaries, and Sets

Day 6: File Handling and Exception Handling

Day 7: Revision and Practice Programs

Students should study for 2–3 hours daily and spend 30 minutes revising previous topics.

Why This Prompt Worked

The prompt includes a specific duration, subject, and target audience. This helps the AI generate a structured and relevant study plan.

3. Creating Interview Questions

Prompt

Generate 10 beginner-level Python interview questions with answers for college students.

AI Response

What is Python?

Python is a high-level, interpreted programming language.

What is a variable?

A variable is used to store data in a program.

What are Python data types?

Common data types include int, float, string, list, and dictionary.

What is a list?

A list is an ordered collection of items.

What is a function?

A function is a reusable block of code.

What is a loop?

A loop repeats a set of instructions.

What is the difference between a list and a tuple?

Lists are mutable, while tuples are immutable.

What is a dictionary?

A dictionary stores data in key-value pairs.

What is indentation in Python?

Indentation defines code blocks in Python.

Why is Python popular?

Python is simple, readable, and versatile.

Why This Prompt Worked

The prompt specifies the topic, number of questions, difficulty level, and requirement for answers, resulting in a useful interview preparation set.

4. Writing a Resume Summary

Prompt

Write a professional resume summary for a student with skills in Python programming, AI fundamentals, GitHub, and problem-solving.

AI Response

Motivated and enthusiastic student with a strong foundation in Python programming, Artificial Intelligence fundamentals, and GitHub version control. Possesses excellent problem-solving abilities and a passion for learning new technologies. Seeking opportunities to apply technical knowledge and contribute to innovative projects while continuing professional growth.

Why This Prompt Worked

The prompt provides the student's skills and desired outcome, enabling the AI to create a professional and focused resume summary.

5. Creating a Lesson Plan

Prompt

Create a one-hour lesson plan to teach Python basics to beginners. Include objectives, activities, and assessment.

AI Response

Topic: Introduction to Python Programming

Duration: 1 Hour

Objectives:

Understand Python basics.

Learn variables and data types.

Write simple Python programs.

Activities:

Introduction to Python (10 minutes)

Variables and Data Types (15 minutes)

Practical Coding Session (20 minutes)

Question and Answer Session (10 minutes)

Quick Quiz (5 minutes)

Assessment: Students will create a simple Python program that displays their name and age.